

Bishop's University
Department of Computer Science
CS504 – Assignment 1

Objective: Linear Regression in Data Science (Python and R)

Learn how to implement linear regression, including preprocessing steps, in both Python and R. This exercise will strengthen your understanding of data preparation, model implementation, and comparison of the results across two different programming languages.

Problem Statement:

You are provided with a dataset (**California Housing Dataset**) containing information about house prices. The dataset includes various features (see description in page 3). Your task is to predict the price of a house based on these features using linear regression.

The first 5 rows of the dataset are shown in the following table:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

Tasks:

1. Import required libraries
2. Load the California Housing Dataset
3. Display the first 5 rows of the dataset (see table above)

Part 1: Data Preprocessing

4. Check for missing data (if any)
5. Handle missing data (Impute with mean values)
6. Feature scaling using **StandardScaler**

Part 2: Linear Regression Model

7. Split the dataset into train and test sets

8. Train a linear regression model
9. Predict on the test set.
10. Evaluate the model on the testing set and calculate the R^2 score and Mean Squared Error (MSE).
11. Display learned weights for each feature and bias

Part 3: Visual Analysis

12. Visualize the relationship between predicted and actual house prices.

Libraries:

- **Python:** matplotlib, pandas
 - **R:** dplyr, ggplot2, caret
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Submission:

- 1) Submit your codes in both Python and R (2 separate Jupyter Notebooks).
- 2) At the top of the notebook, include the following text:

▼ Assignment 1

Name: your full name

Code: your ID

- 3) You must answer each question (task) in one SINGLE code cell (NO MORE, NO LESS), including short comments to explain each step.
- 4) Label each task clearly before its corresponding code cell, as follows:

▼ 1. Import necessary libraries

```
1 import numpy as np
2 import pandas as pd
3 ....
```

▼ 2. Load the California Housing Dataset

```
1 california_data = fetch_california_housing()
```

Dataset description

The **California Housing Dataset** is a popular dataset used in machine learning for regression tasks. It contains data collected during the 1990 U.S. Census and provides information about housing in California districts. The goal is to predict the median house value based on several features.

Key Features:

- **MedInc (Median Income):** Median income in the district, scaled between 0 and 15 (in tens of thousands of dollars).
- **HouseAge:** Median age of the houses in the district.
- **AveRooms:** Average number of rooms per household.
- **AveBedrms:** Average number of bedrooms per household.
- **Population:** Population of the district.
- **AveOccup:** Average number of household members.
- **Latitude:** Geographical latitude of the district.
- **Longitude:** Geographical longitude of the district.

Target:

- **MedHouseVal (Median House Value):** The median house value in the district, the variable you aim to predict, in thousands of dollars.

Purpose:

The dataset is primarily used to predict **median house values** based on the socioeconomic and geographic features of the districts. It is well-suited for linear regression, requiring preprocessing such as handling missing data (e.g., in bedroom counts), scaling features (e.g., income), and cleaning outliers.

Size:

- **Rows:** 20,640 samples (each representing a district in California).
- **Columns:** 8 features + target (house value).

Source: https://scikit-learn.org/stable/datasets/real_world.html#california-housing-dataset